

**The Role of Beta in Shaping the Relationship Between Dividend Payout Ratios and
Price-to-Earnings Valuation (2018–2023)**

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June 30, 2026

Abstract

This study examines the relationship between dividend payout ratios and price-to-earnings (P/E) ratios among large-cap U.S. companies from 2018 to 2023. It also investigates whether market risk, measured by beta, influences this relationship. Data from eight companies across low-beta and high-beta groups were analyzed. Companies were classified based on their average beta values, with beta calculated using annual stock returns over the study period. To measure central tendencies, beta values were averaged, while dividend payout ratios and P/E ratios were analyzed using median values to reduce the impact of outliers, particularly from Ford's data. Results show that the low-beta group had higher dividend payout ratios and P/E ratios compared to the high-beta group. Regression analysis between dividend payout ratios and P/E ratios produced an R^2 value of 0.004, indicating that dividend payout ratios alone do not explain significant differences in P/E ratios among large-cap U.S. companies. Overall, the findings suggest that while beta groups show differences in dividend behavior and valuation, dividend payout ratios alone have a limited effect on P/E ratios.

Keywords: Dividend payout ratios, P/E ratios, market risk, beta

The Role of Beta in Shaping the Relationship Between Dividend Payout Ratios and Price-to-Earnings Valuation (2018–2023)

Stock valuation is an important part of understanding how investors assess the worth of a company, and it is often measured using the price-to-earnings (P/E) ratio. Another key factor in corporate finance is dividend policy, commonly measured through the dividend payout ratio, which shows how much of a company's earnings are returned to shareholders. In addition, market risk, measured by beta, reflects how sensitive a company's stock is to movements in the overall market and may influence both valuation and dividend decisions. These measures are important because they help investors compare companies and make decisions about risk and return across different sectors. This study examines the relationship between dividend payout ratios and P/E ratios among large-cap U.S. companies from 2018 to 2023 and investigates whether this relationship differs between low-beta and high-beta firms.

Method

Participants

The study analyzed eight large-cap U.S. companies from 2018 to 2023. The selected companies represented four industries: Food and Beverage, Consumer Packaged Goods, Automobiles and Auto Components, and Technology. Companies were selected to represent different levels of market risk and dividend policies. The companies were divided into low-beta and high-beta groups based on their calculated average beta values during the study period.

Assessments and Measures

Beta

Beta was used as the measurement of market risk. Annual beta values were calculated using a three-year rolling window by comparing each company's stock returns to S&P 500

market returns over the study period from 2018 to 2023. Each rolling beta value was calculated using the previous three years of data, and the resulting annual beta values were then averaged to produce a single beta value for each company. The average beta for each company was used for classification. Companies with beta values greater than 0.9 were classified as high beta, while companies with beta values of 0.9 or lower were classified as low beta. The 0.9 benchmark was selected because market conditions during the COVID-19 period caused significant shifts in beta behavior across industries, as shown in research discussing the COVID-19 beta effect.

Dividend Payout Ratio

Dividend payout ratio was used to measure how much of a company's earnings were distributed to shareholders as dividends. It was calculated using:

$$\text{Dividend Payout Ratio} = \text{Dividends Per Share} / \text{Earnings Per Share}$$

Annual dividend payout ratios were calculated for each company from 2018 to 2023.

Median values were used in the final analysis to reduce the impact of outliers.

Price-to-Earnings Ratio

The P/E ratio was used as the measurement of company valuation. It was calculated using:

$$\text{P/E Ratio} = \text{Market Price Per Share} / \text{Earnings Per Share}$$

Year-end stock prices and earnings per share values were used to calculate annual P/E ratios. Median P/E ratios were used for final comparisons between groups.

Data Collection

Financial data, such as dividend payments and earnings per share values, was collected from company annual reports and 10-K filings. Historical stock prices and S&P 500 returns were collected from Stooq to calculate yearly beta values.

Procedure and Analysis

Annual beta, dividend payout ratio, and P/E ratio values were calculated for each company from 2018 to 2023. Company-level averages and medians were then calculated and used for comparison between low-beta and high-beta groups. A regression analysis was conducted between dividend payout ratios and P/E ratios to evaluate whether dividend payout ratios had a relationship with valuation differences among the selected companies.

Results

This section presents the findings from the comparison of low-beta and high-beta companies, as well as the relationship between dividend payout ratios and price-to-earnings (P/E) ratios.

Outcome 1: Beta Group Differences

The analysis showed clear differences between low-beta and high-beta companies. Low-beta firms had an average beta of 0.522, compared to 1.100 for high-beta firms. Low-beta companies also had higher median dividend payout ratios (0.701) compared to high-beta companies (0.372). In terms of valuation, low-beta firms had a higher median P/E ratio of 27.54, while high-beta firms had a lower median P/E ratio of 19.94.

Outcome 2: Relationship Between Dividend Payout Ratio and P/E Ratio

A regression analysis was conducted to test the relationship between dividend payout ratios and P/E ratios across all companies. The results showed an extremely weak relationship, with an R^2 value of 0.004. This indicates that dividend payout ratios explain very little of the variation in P/E ratios among the selected companies.

Discussion

Limitations. This study has several limitations that may affect the interpretation of the results. First, the sample size is relatively small, with only eight large-cap U.S. companies, which limits how broadly the findings can be applied to the overall market. Second, the analysis only includes firms from selected industries, meaning the results may be influenced by sector-specific trends rather than general market behavior. Third, beta was calculated using a three-year rolling average, which helps smooth short-term volatility but may also reduce sensitivity to sudden market changes, especially during unusual periods such as COVID-19. In addition, the study

only uses dividend payout ratios and beta as explanatory variables for P/E ratios, meaning other important factors such as earnings growth, interest rates, and industry conditions are not included. These limitations suggest that while the results show clear patterns within the sample, they should not be interpreted as fully generalizable to all market conditions.

The results suggest that market risk, measured by beta, is associated with differences in both dividend policy and valuation levels. Low-beta companies tended to have higher dividend payout ratios and higher P/E ratios, suggesting that investors may place a premium on more stable, lower-risk firms.

However, the relationship between dividend payout ratios and P/E ratios alone was extremely weak. This implies that dividend policy by itself is not a strong predictor of valuation across large-cap companies. Instead, valuation appears to be influenced more strongly by broader factors such as industry characteristics, earnings stability, and overall market risk.

Overall, the findings suggest that beta plays a more important role than dividend payout ratios in explaining differences in valuation among large-cap U.S. companies during the 2018–2023 period.

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